Graph Valid Tree (medium)

Problem Statement

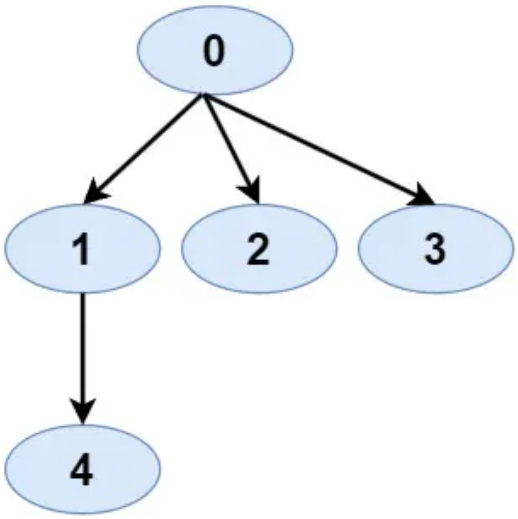
Given a number `n`, which indicates the number of nodes numbered from `0` to `n-1`, and a list of undirected edges for the graph, determine if the graph is a valid tree.

A graph qualifies as a valid tree if it meets the following criteria:

1. It has no cycles.
2. It is fully connected.

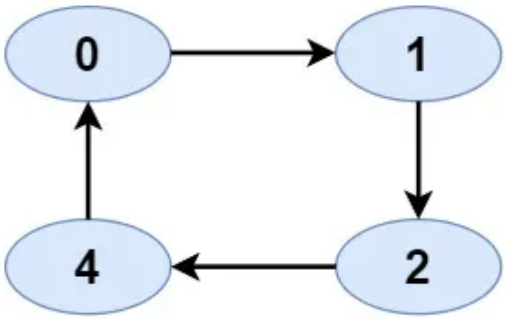
Examples

- **Example 1**:
  - **Input**: n = 5, edges = [[0,1],[0,2],[0,3],[1,4]]

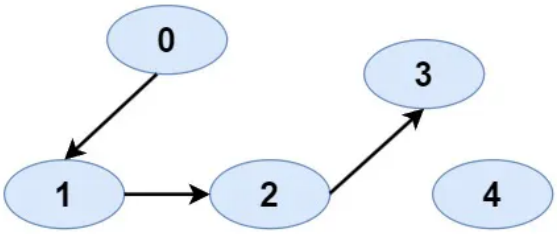


- **Expected Output**: true
- **Justification**: There are no cycles in the graph, and all nodes are connected, forming a valid tree.

- **Example 2**:
  - **Input**: n = 4, edges = [[0,1],[1,2],[2,3],[3,0]]]]



- **Expected Output**: false
- **Justification**: There is a cycle in the graph (0-1-2-3-0), thus it's not a valid tree.
- **Example 3**:
  - **Input**: n = 5, edges = [[0,1],[1,2],[2,3]]



- **Expected Output**: false
- **Justification**: TNode 4 is not connected to any other node, making the graph not fully connected.

Try it yourself

Try solving this question here:

Java

```java
import java.util.*;

public class Solution {
    public boolean validTree(int n, int[][] edges) {
        // ToDo: Write Your Code Here.
        return false;
    }
```

```
8     }
9
```

Test Case | Run Code Result | Stdout

```
5
[[0,1],[0,2],[0,3],[1,4]]
```

Use Example Testcases | Run | Submit

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Solution: Number Of Connected Components In An Undirected Graph

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Solution: Graph Valid Tree

Mark as Completed